Exercise Sheet 5

Discrete Mathematics by Hongfei Fu, 2023.9.28

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Note:The following exercises involve the logical operators $NAND$ and NOR. The proposition $p\ NAND\ q$ is true when either p or q, or both, are false; and it is false when both p and q are true. The proposition $p\ NOR\ q$ is true when both p and q are false, and it is false otherwise. The propositions $p\ NAND\ q$ and $p\ NOR\ q$ are denoted by $p|q$ and $p \downarrow q$, respectively. (The operators — and $\downarrow$ are called the Sheffer stroke and the Peirce arrow after H. M. Sheffer and C. S. Peirce, respectively.)

1. In this exercise we will show that $\{\downarrow\}$ is a functionally complete collection of logical operators.

   a) Show that $p \downarrow p$ is logically equivalent to $\neg p$.

   b) Show that $(p \downarrow q) \downarrow (p \downarrow q)$ is logically equivalent to $p \vee q$.

   c) Conclude from parts (a) and (b), and Exercise 49 , that $\{\downarrow\}$ is a functionally complete collection of logical operators.

   Note::Show that $p \downarrow q$ is logically equivalent to $\neg(p \vee q)$

   Answer Area:

   a)

p	$\neg p$	$p \downarrow p$
T	F	F
F	T	T

   b)

p	q	$(p \downarrow q) \downarrow (p \downarrow q)$	$p \vee q$
T	T	T	T
F	T	T	T
T	F	T	T
F	F	F	F

   c) $\because \neg p \equiv p \downarrow p, p \vee q \equiv (p \downarrow q) \downarrow (p \downarrow q)$, and the collection $\neg, \vee$ is functionally complete.
   $\therefore$ Any compound proposition is logically equivalent to a compound proposition that involves only the logical operators $\neg$ and $\vee$.Then we can use $\downarrow$ to replace every occurrence of $\neg$ and $\vee$.So any compound proposition is logically equivalent to a compound proposition that involves only the logical operators $\downarrow$.

2. Show that $p|(q|r)$ and $(p|q)|r$ are not equivalent, so that the logical operator $|$ is not associative.
   Answer Area:
   Suppose that:

p	q	r	$p\|(q\|r)$	(p — q) — r
T	T	F	F	T

   $\therefore p|(q|r)$ and $(p|q)|r$ are not equivalent,so that the logical operator $|$ is not associative.

3. Put these statements in prenex normal form, where the domain of all variables are the same and nonempty. [Hint: Use logical equivalence from Tables 6 and 7 in Section 1.3, Table 2 in Section 1.4, Example 19 in Section 1.4, Exercises 45 and 46 in Section 1.4, and Exercises 48 and 49.]
b) $\neg(\forall x P(x) \vee \forall x Q(x))$
c) $(\exists x P(x)) \rightarrow (\exists x Q(x))$

Answer Area:
b) $\neg(\forall x P(x) \vee \forall x Q(x)) \equiv \neg(\forall x P(x)) \wedge \neg(\forall x Q(x))$
$\equiv (\exists x \neg P(x)) \wedge (\exists x \neg Q(x))$
$\equiv \exists x \exists y(\neg P(x) \wedge \neg Q(x))$
c) $(\exists x P(x)) \rightarrow (\exists x Q(x)) \equiv \neg(\exists x P(x)) \vee (\exists x Q(x))$
$\equiv (\forall x \neg P(x)) \vee (\exists x Q(x))$
$\equiv \forall x \exists y(\neg P(x) \vee Q(x))$

4. Consider the compound proposition $\phi = p \rightarrow (q \oplus r)$ where p, q, r are propositional variables.

(a) Give a disjunctive normal form of ϕ.

(b) Give a conjunctive normal form of ϕ.

Answer Area:

(a)

p	q	r	$p \rightarrow (q \oplus r)$
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	F
F	T	T	T
F	T	F	T
F	F	T	T
F	F	F	T

Then we have $\phi \equiv (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r)$

(b)

p	q	r	$\neg(p \rightarrow (q \oplus r))$
T	T	T	T
T	T	F	F
T	F	T	F
T	F	F	T
F	T	T	F
F	T	F	F
F	F	T	F
F	F	F	F

Then we have $\neg\phi \equiv (p \wedge q \wedge r) \vee (p \wedge \neg q \wedge \neg r)$
$\phi \equiv \neg((p \wedge q \wedge r) \vee (p \wedge \neg q \wedge \neg r)) \equiv (\neg p \vee \neg q \vee \neg r) \wedge (\neg p \vee q \vee r)$